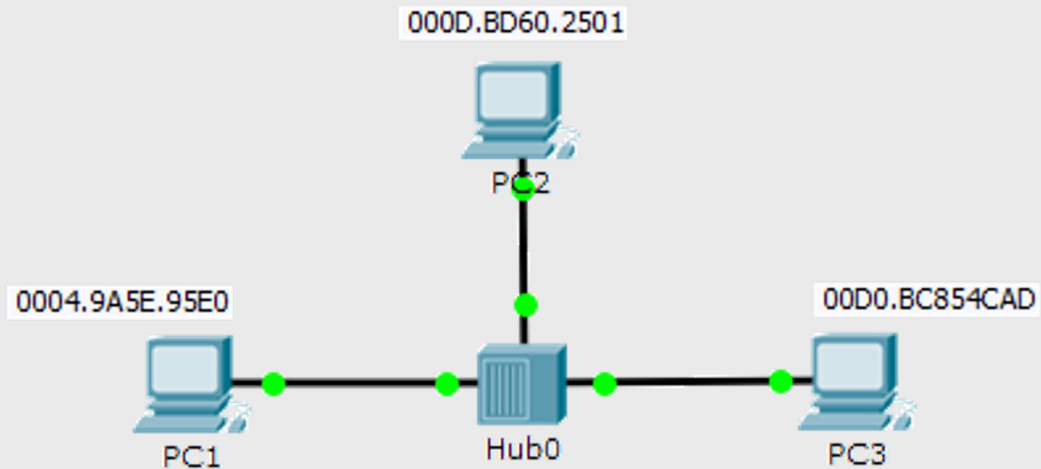


- Ethernet LANs
 - **Spanning Tree Protocol**
 - **First hop redundancy**
 - **Link aggregation**
 - **Controllo degli accessi**
 - **Virtual LAN**
- Gestione avanzata degli indirizzi
 - **Network Address Translation**
 - **Dynamic Host Configuration Protocol**
- Wide Area Networks

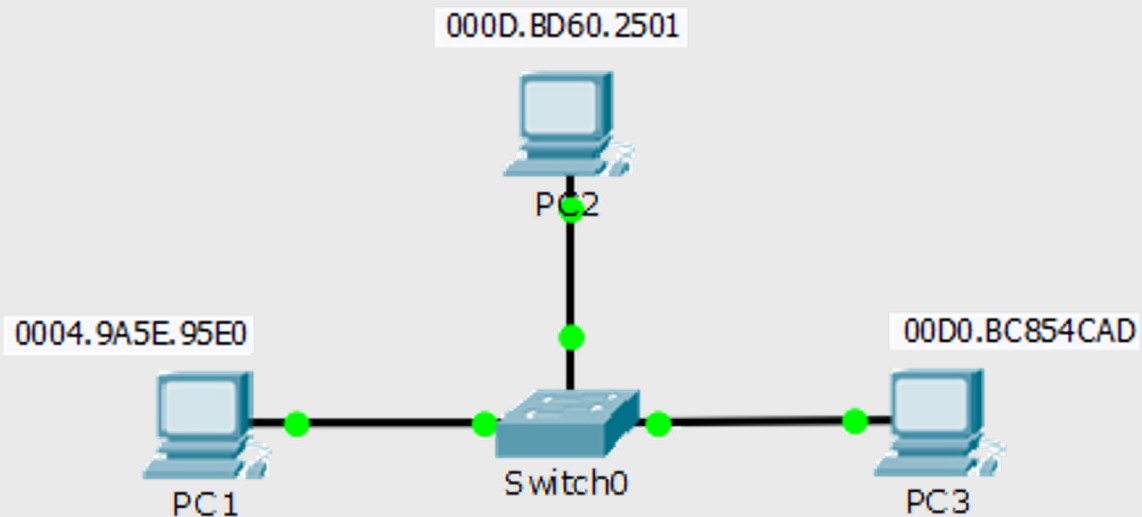
Lab8

LAN Switching
Basic IOS switch configuration

LAN switching recap



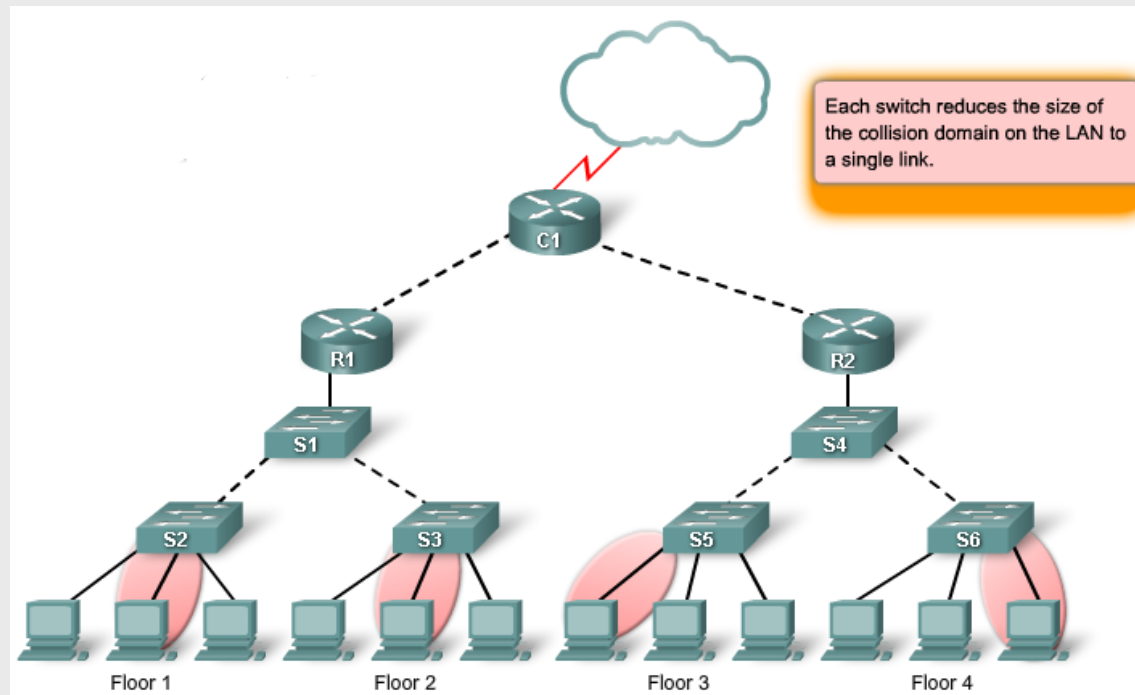
**collision domain
=
broadcast domain**



**collision domain
≠
broadcast domain**

LAN switching recap

- Switched Ethernet (transparent bridging)



LAN switching recap

- Ethernet physical layer ([Wikipedia](#))

Ethernet Name	Cable Type	Maximum Speed	Maximum Transmission Distance	Cable Name
100Base-TX	UTP	100Mbps	100 Meters	CAT5, CAT5e, CAT6
1000Base-T	UTP	1000Mbps	100 Meters	CAT5e, CAT6
1000Base-SX	Fiber	1000Mbps	550 Meters	Multimode and Singlemode Fiber
1000Base-LX	Fiber	1000Mbps	550 Mbps MMF, 2000 Meters SMF	Singlemode Fiber
1000Base-ZX	Fiber	1000Mbps	70000 Meters (70 Kilometers)	Singlemode Fiber
10GBase-T	UTP	10Gbps	100 Meters	CAT5e, CAT6
10GBase-SR	Fiber	10Gbps	300 Meters	Multimode Fiber
10GBase-LR	Fiber	10Gbps	10000 Meters (10 Kilometers)	Singlemode Fiber
10GBase-ER	Fiber	10Gbps	40000 Meters (40 Kilometers)	Singlemode Fiber
10GBase-SW	Fiber	10Gbps	300 Meters	Multimode Fiber
10GBase-LW	Fiber	10Gbps	10000 Meters (10 Kilometers)	Singlemode Fiber
10GBase-EW	Fiber	10Gbps	40000 Meters (40 Kilometers)	Singlemode Fiber

Multimode Fiber



Singlemode Fiber



10G Multimode Fiber



SFP+Copper (Twinax)



ITMPS.COM

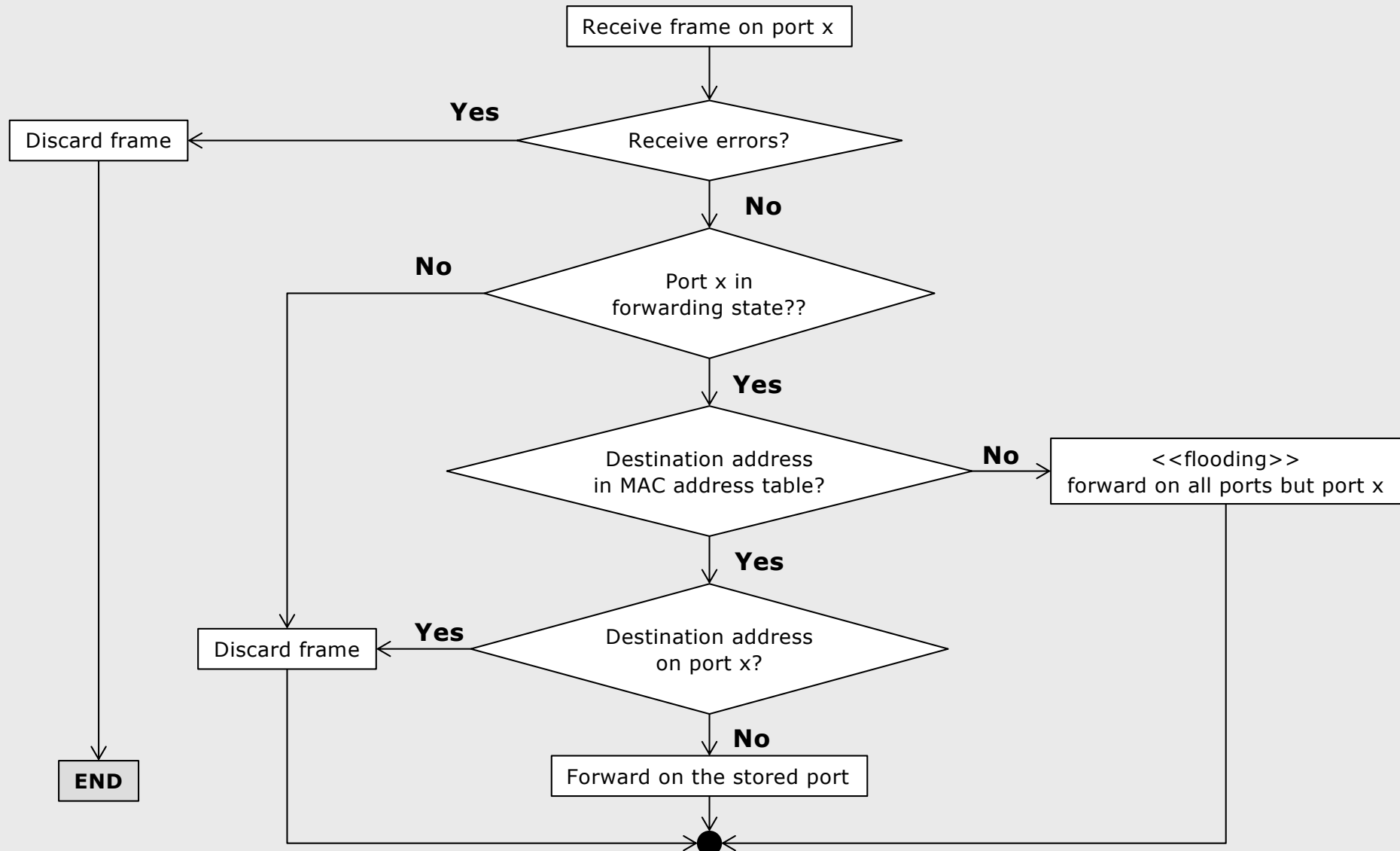
LAN switching recap

- MAC address database
 - Is the forwarding table of the switch
- Entry data
 - MAC address
 - Port id
 - Vlan id
 - Type
 - Static
 - Dynamic
 - Ageing
 - STP state

```
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
-----
All     0100.0ccc.cccc   STATIC    CPU
All     0100.0ccc.cccd   STATIC    CPU
All     0180.c200.0000   STATIC    CPU
All     0180.c200.0001   STATIC    CPU
All     0180.c200.0002   STATIC    CPU
All     0180.c200.0003   STATIC    CPU
All     0180.c200.0004   STATIC    CPU
All     0180.c200.0005   STATIC    CPU
All     0180.c200.0006   STATIC    CPU
All     0180.c200.0007   STATIC    CPU
All     0180.c200.0008   STATIC    CPU
All     0180.c200.0009   STATIC    CPU
All     0180.c200.000a   STATIC    CPU
All     0180.c20000d     STATIC    CPU
All     0180.c200.000e   STATIC    CPU
All     0180.c200.000f   STATIC    CPU
All     0180.c200.0010   STATIC    CPU
All     ffff.ffff.ffff   STATIC    CPU
1       000c.7671.7534   DYNAMIC    Fa0/2
1       0013.e809.7695   DYNAMIC    Fa0/2
1       0017.9a51.d339   DYNAMIC    Fa0/2
1       0019.5b0a.a951   DYNAMIC    Fa0/2
1       0060.b0af.7be4   DYNAMIC    Fa0/2
Total Mac Addresses for this criterion: 25
```

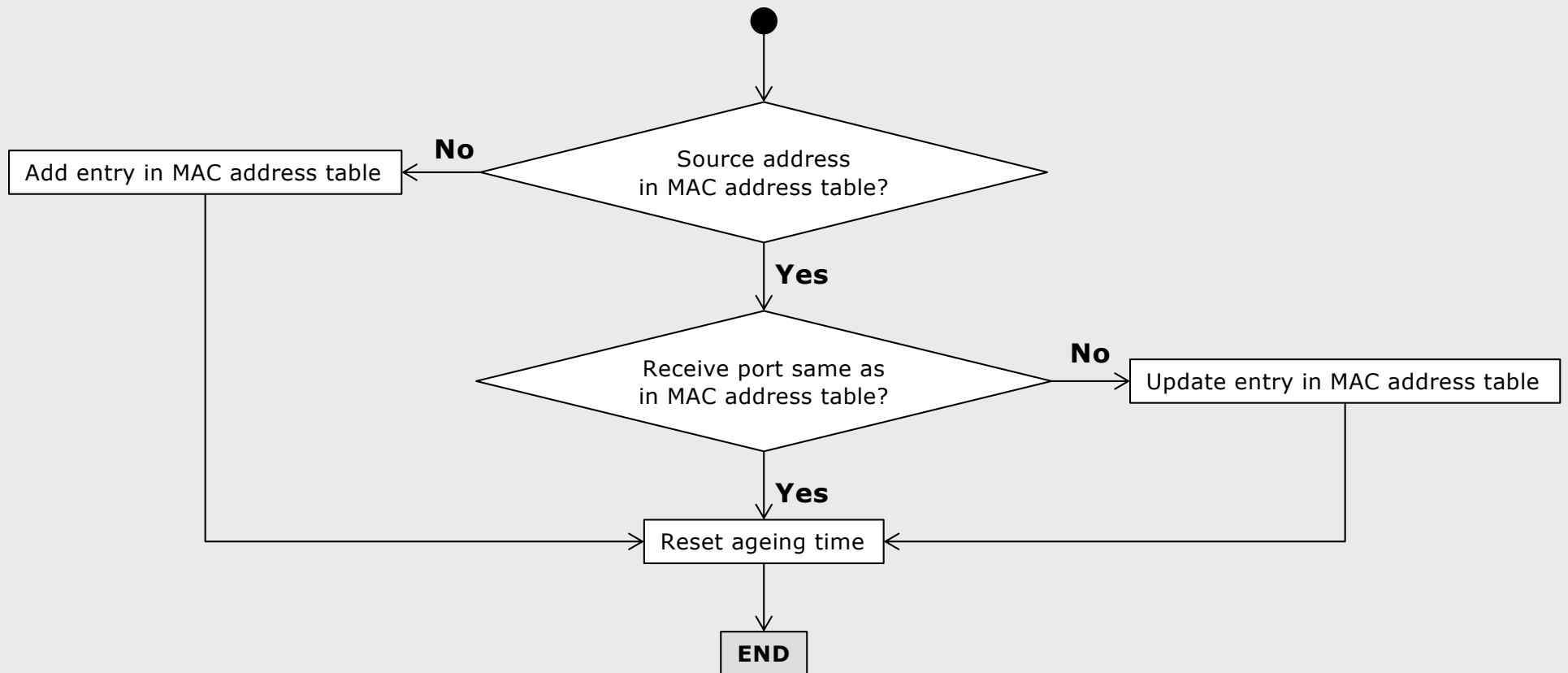
LAN switching recap

■ Forwarding process

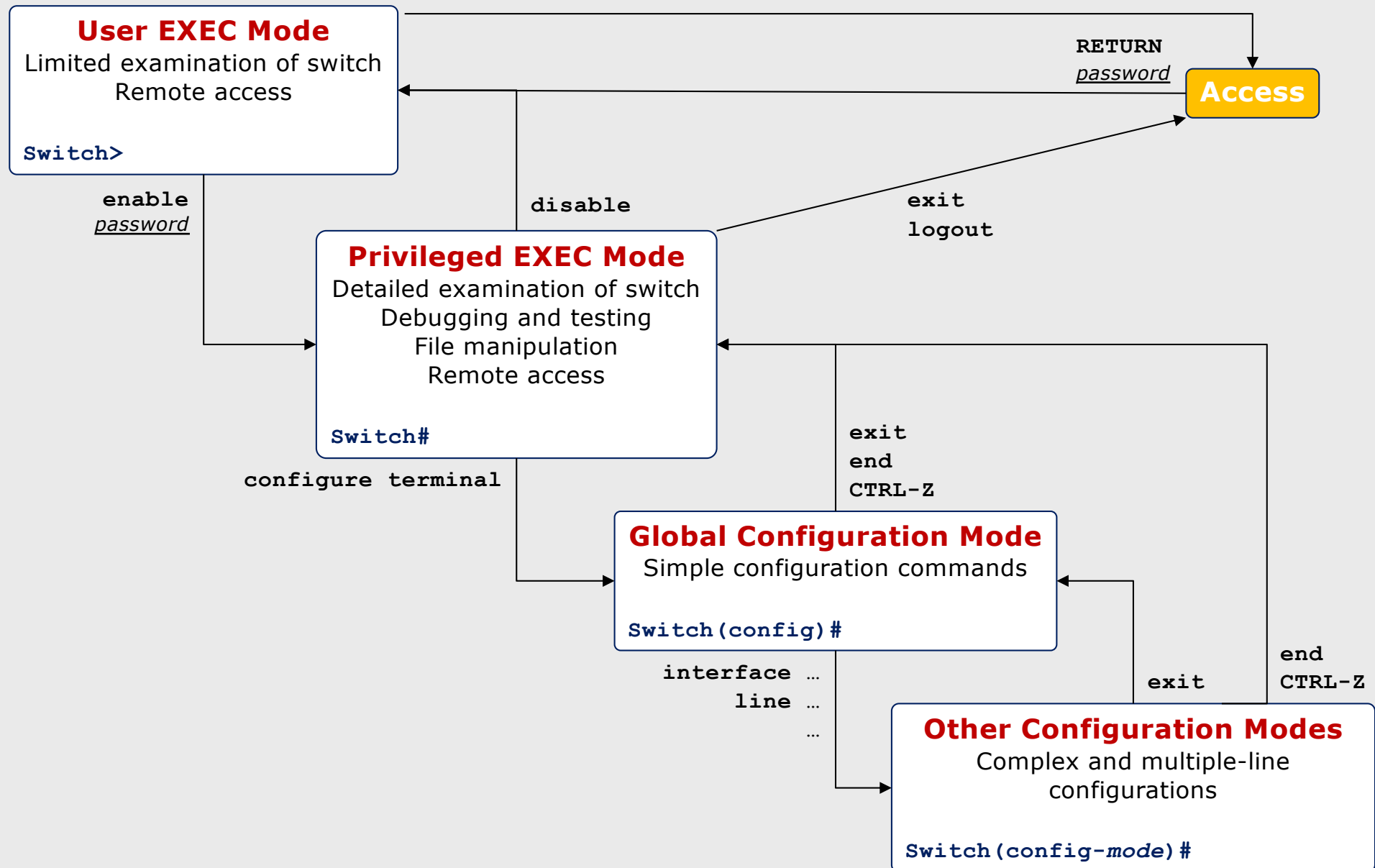


LAN switching recap

■ Learning process

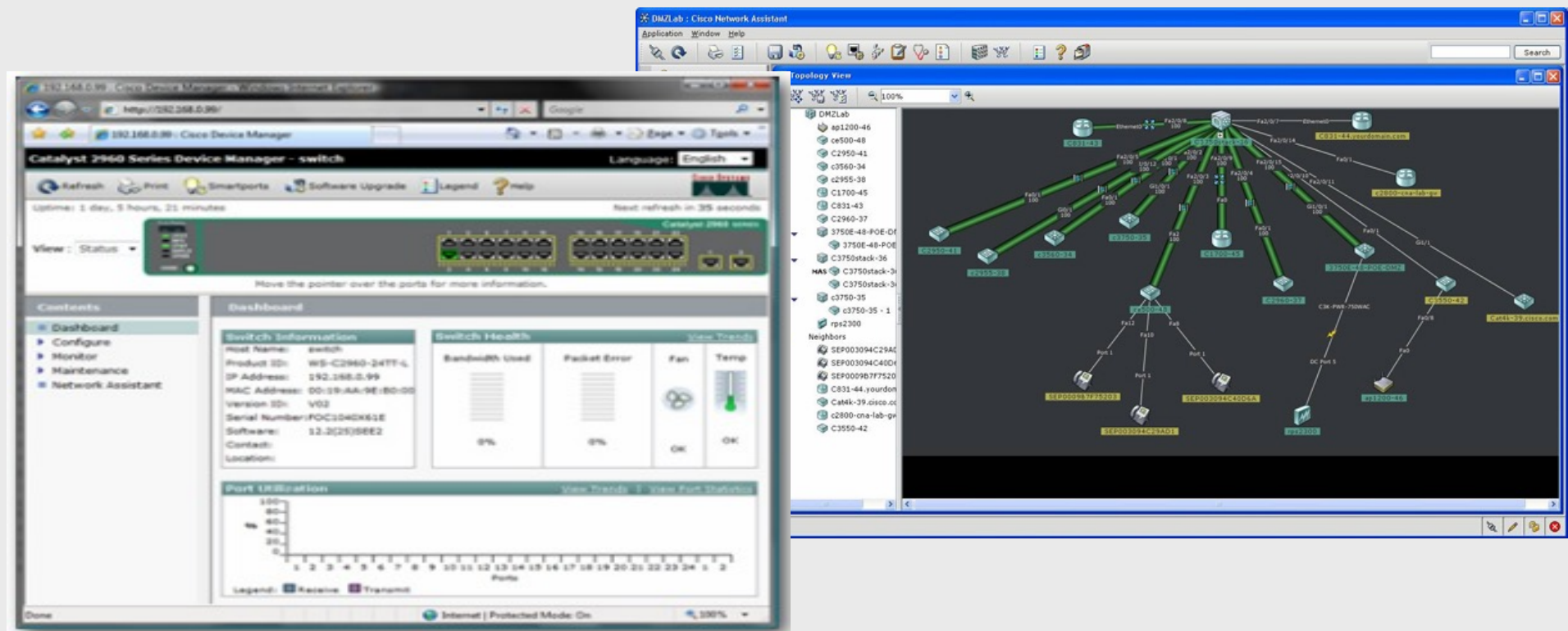


Switch management configuration (IOS)



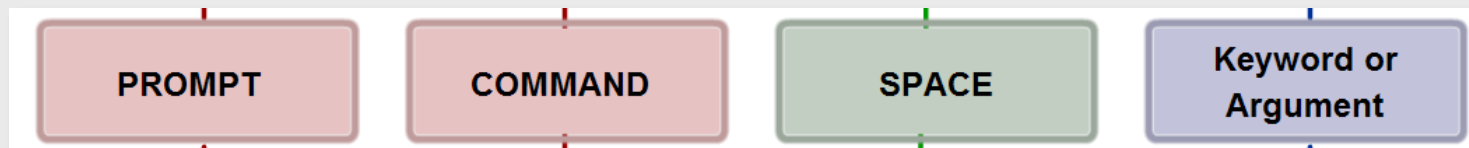
Switch management configuration (IOS)

- GUI-based alternatives exist (usually require IP reachability)
 - Network Management Applications
 - Cisco Network Assistant
 - Cisco View (SNMP-based)
 - ...
 - Web-based device built-in configuration software



Switch management configuration (IOS)

■ IOS command structure



■ Context-sensitive help

Example of a sequence of commands using the CLI context sensitive help

```
Cisco#cl?  
clear clock  
Cisco#clock ?  
    set Set the time and date  
Cisco#clock set  
% Incomplete command.  
Cisco#clock set ?  
    hh:mm:ss Current Time  
Cisco#clock set 19:50:00  
% Incomplete command.
```

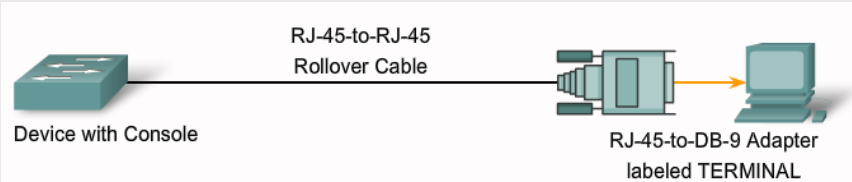
Command explanations
Incomplete Command messages
Invalid input messages
Variable formats

```
Cisco#clock set 19:50:00 ?  
    <1-31> Day of the month  
    MONTH Month of the year  
Cisco#clock set 19:50:00 25 6  
                                ^  
Invalid input detected at '^' marker.  
Cisco#clock set 19:50:00 25 June  
% Incomplete command.  
Cisco#clock set 19:50:00 25 June ?  
    <1993-2035> Year  
Cisco#clock set 19:50:00 25 June 2007  
Cisco#
```

Switch boot-up sequence

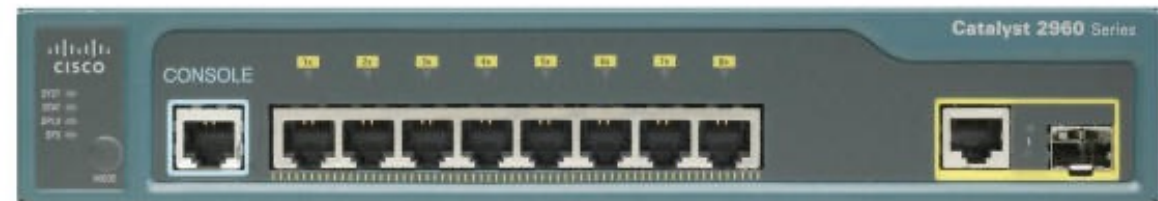
- Load the boot-loader from ROM
- Boot-loader
 - Perform low-level CPU initialization
 - Perform POST (Power-On Self-Test)
 - Initialize the flash file system
 - Load a default operating system software image into memory and boots the switch
 - Default rules to locate the Cisco IOS image
- IOS
 - Initialize the switch executing commands

Switch boot-up sequence



```
Copyright (c) 1986-2006 by Cisco Systems, Inc.  
Compiled Fri 28-Jul-06 04:33 by yenh  
Image text-base: 0x00003000, data-base: 0x00AA2F34  
flashfs[1]: 602 files, 19 directories  
flashfs[1]: 0 orphaned files, 0 orphaned directories  
flashfs[1]: Total bytes: 32514048  
flashfs[1]: Bytes used: 7715328  
flashfs[1]: Bytes available: 24798720  
flashfs[1]: flashfs fsck took 1 seconds.  
flashfs[1]: Initialization complete....don  
flashfs.
```

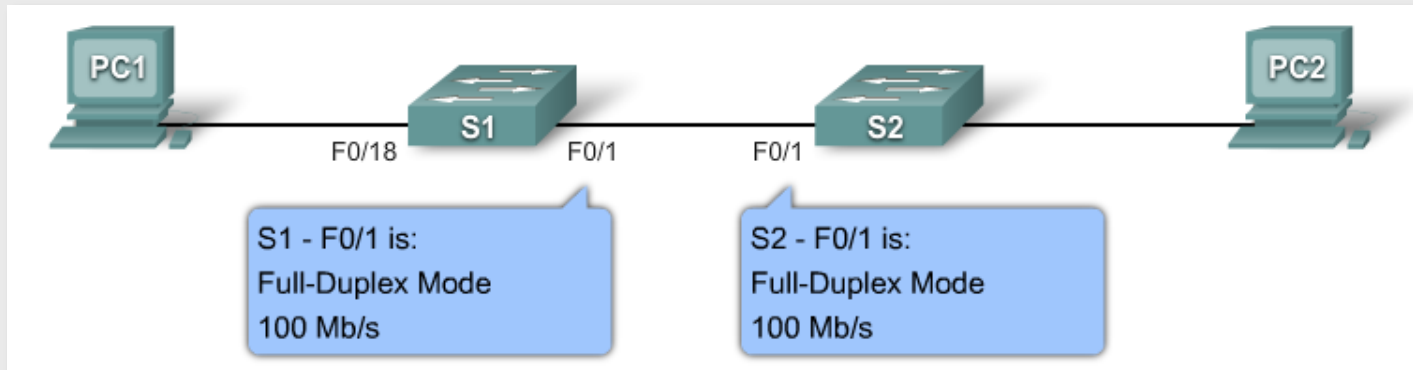
```
POST: CPU MIC register Tests : Begin  
POST: CPU MIC register Tests : End, Status Passed  
  
POST: PortASIC Memory Tests : Begin  
POST: PortASIC Memory Tests : End, Status Passed  
  
POST: CPU MIC PortASIC interface Loopback Tests : Begin  
POST: CPU MIC PortASIC interface Loopback Tests : End, Status  
Passed  
  
POST: PortASIC RingLoopback Tests : Begin  
POST: PortASIC RingLoopback Tests : End, Status Passed  
  
POST: PortASIC CAM Subsystem Tests : Begin
```



POST ok: SYST LED rapidly blinks green
POST fails: SYST LED turns amber

Basic configuration

- Configure duplex and speed, enable automatic MDI-crossover



```
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#interface Fa0/1
S1(config-if)#duplex ?
    auto  Enable AUTO duplex configuration
    full  Force full duplex operation
    half  Force half-duplex operation
S1(config-if)#speed ?
    10    Force 10 Mbps operation
    100   Force 100 Mbps operation
    auto  Enable AUTO speed configuration
S1(config-if)#mdix ?
    auto  Enable automatic MDI crossover detection on this interface
S1(config-if)#end
S1#
```

Basic configuration

■ Manage the MAC Address Table

```
Switch#show mac-address-table
```

Mac Address Table			

Vlan	Mac Address	Type	Ports
----	-----	-----	-----
99	0003.e4ea.0b02	DYNAMIC	Fa0/5
99	00d0.baed.1acb	DYNAMIC	Fa0/18

■ Configure a static MAC address (not aged out)

```
Switch(config)#mac-address-table static mac_address vlan vlan-id interface-id
```

■ Modify the default aging time (300 s)

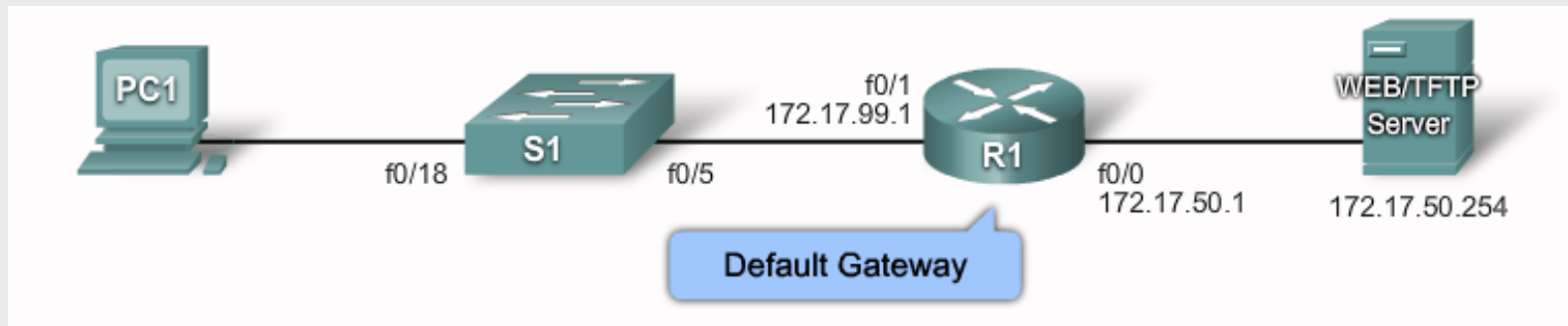
```
Switch(config)#mac-address-table aging-time seconds [vlan vlan_id]
```

Verifying configuration

Cisco IOS CLI Command Syntax	
Displays interface status and configuration for a single or all interfaces available on the switch.	<code>show interfaces [interface-id]</code>
Displays contents of startup configuration.	<code>show startup-config</code>
Displays current operating configuration.	<code>show running-config</code>
Displays information about flash: file system.	<code>show flash:</code>
Displays system hardware and software status.	<code>show version</code>
Display the session command history.	<code>show history</code>
Displays IP information. The interface option displays IP interface status and configuration. The http option displays HTTP information about device manager running on the switch. The arp option displays the IP ARP table.	<code>show ip {interface http arp}</code>
Displays the MAC forwarding table.	<code>show mac-address-table</code>

Assign IP address (basic)

- Enable remote configuration using TCP/IP
 - Assign the switch an IP address



```
S1#configure terminal
S1(config)#interface vlan 1
S1(config-if)#ip address 172.17.99.2 255.255.255.0
S1(config-if)#no shutdown
S1(config-if)#exit
S1#ip default-gateway 172.17.99.1
S1#end
```

Configuration management

■ Backup and restore *switch* configurations

```
SW-lab#copy running-config startup-config
```

```
SW-lab#copy startup-config flash:
```

```
Destination filename [startup-config]? startup-config.bak
```

```
551 bytes copied in 0.416 secs (1324 bytes/sec)
```

```
SW-lab#copy running-config flash:
```

```
Destination filename [running-config]? running-config.bak
```

```
Building configuration...
```

```
[OK]
```

```
SW-lab#copy flash: startup-config
```

```
Source filename []? startup-config.bak
```

```
Destination filename [startup-config]?
```

```
[OK]
```

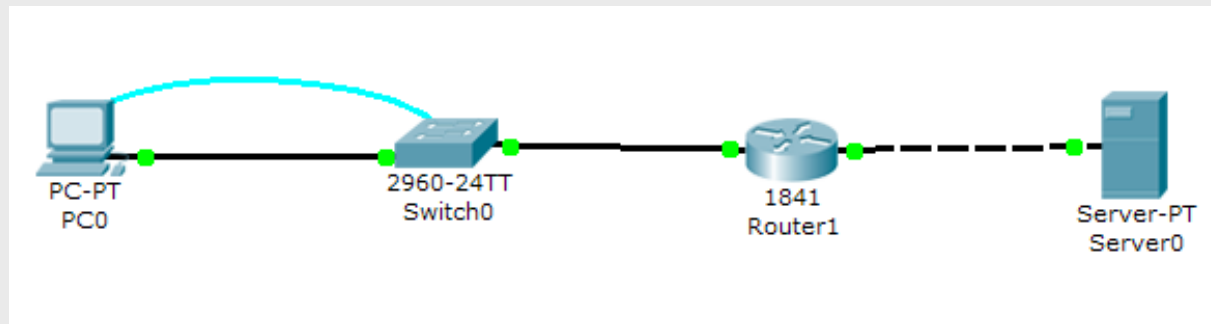
```
551 bytes copied in 0.416 secs (1324 bytes/sec)
```

```
SW-lab#reload
```

```
Proceed with reload? [confirm]
```

Configuration management

■ Backup and restore on the network



```
Switch0#copy running-config tftp:  
Address or name of remote host []?  
Destination filename [Switch-config]?  
  
Switch#copy startup-config tftp:
```

```
Switch0#copy tftp: startup-config  
...
```

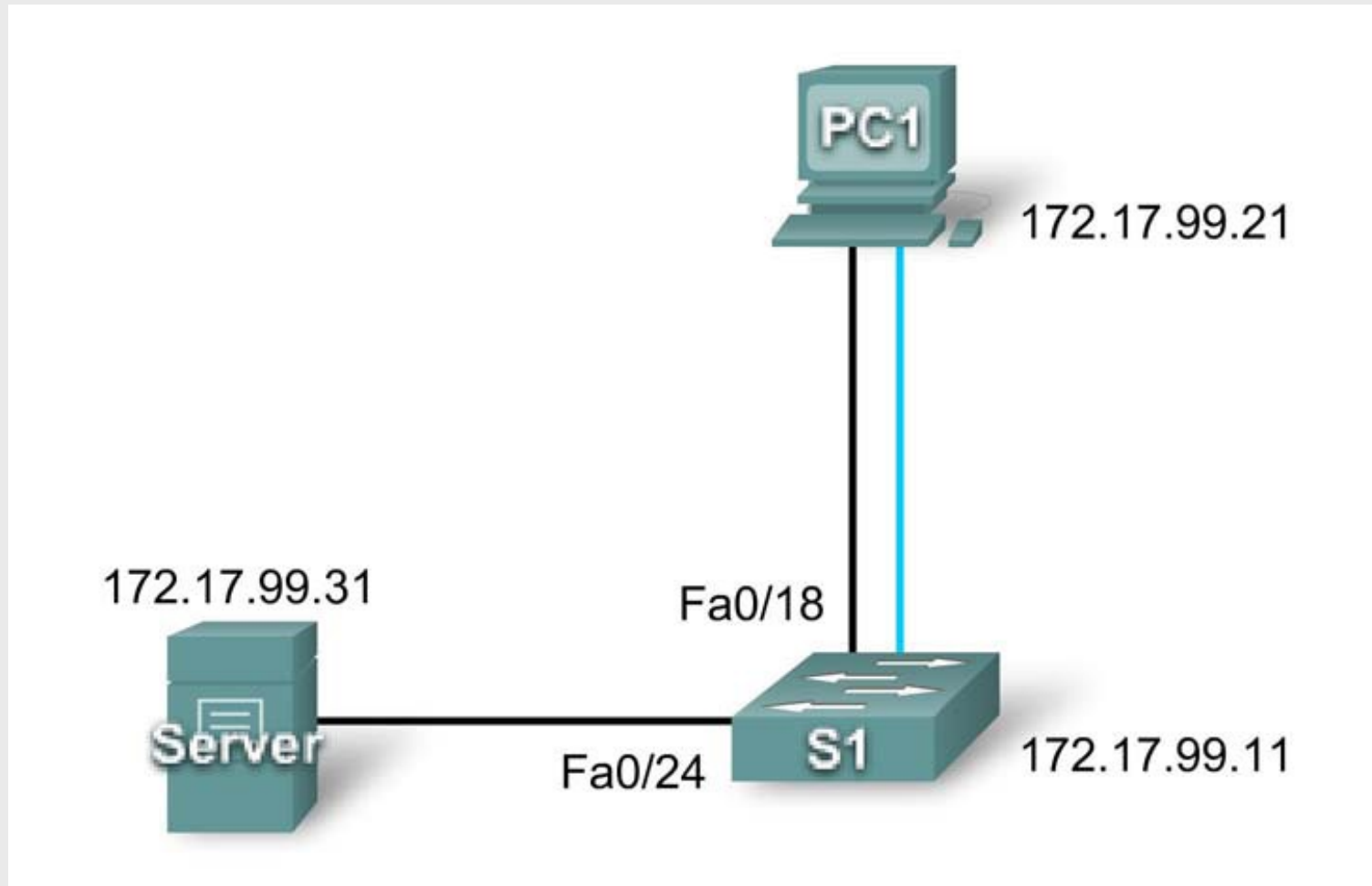
```
Switch0#copy tftp: running-config
```

Attenzione!!!

■ Clearing configuration

```
Switch0#erase startup-config  
  
Switch0#delete flash:  
Delete filename []?
```

Lab activity



Configuring basic switch security

■ Console password

```
Switch(config)#line console 0  
Switch(config-line)#password password  
Switch(config-line)#login
```

■ Virtual Terminal password (telnet)

```
Switch(config)#line vty 0 15  
Switch(config-line)#password password  
Switch(config-line)#login
```

■ Privileged EXEC mode authentication

```
Switch(config)#enable secret password
```

■ Encrypting password display (show commands)

```
Switch(config)#service password-encryption
```

Configuring basic switch security

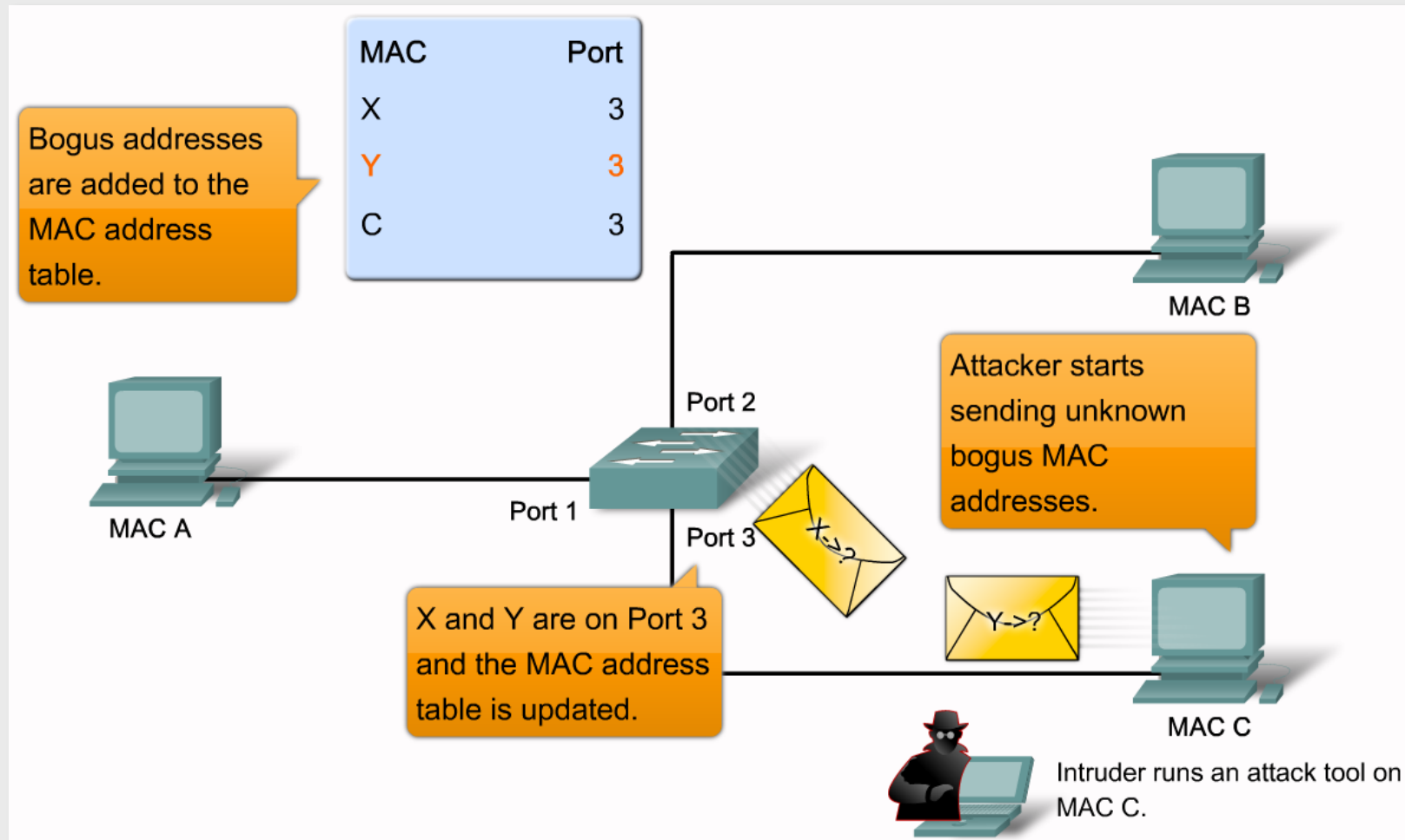
■ Enabling SSH access (instead of telnet)

```
Sw1#conf t
Sw1(config)#ip domain-name mydomain.com
Sw1(config)#crypto key generate rsa
The name for the keys will be: Sw1.mydomain.com
...
How many bits in the modulus [512]: 1024
...
Sw1(config)#end
Sw1#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
Sw1#conf t
Sw1(config)#username admin secret cisco
Sw1(config)#line vty 0 15
Sw1(config-line)#transport input ssh
Sw1(config-line)#login local
Sw1(config-line)#exit
Sw1(config)#ip ssh version 2
Sw1(config)#ip ssh time-out 60
Sw1(config)#ip ssh authentication-retries 5
Sw1(config-line)#^Z
Sw1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 5
```



Switch security: common attacks

■ MAC address flooding



Configuring switch security

- Port security
 - Limit the number of **valid** (secure) MAC addresses allowed on each port
 - Packets with source addresses outside the group of allowed addresses on that port are not forwarded
 - Example: max number = 1, i.e. single secure MAC address assigned
 - The port is reserved for use by the workstation with that particular MAC address
- Secure MAC address types
 - **Static**
 - manually configured (using the `switchport port-security mac-address mac-address` interface configuration command)
 - stored in the address table and saved to the running configuration
 - **Dynamic**
 - dynamically learned
 - stored in the address table only, removed when the switch restarts
 - **Sticky**
 - dynamically learned
 - stored in the address table and saved to the running configuration

Configuring switch security

■ Configure port security

```
Sw1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Sw1(config)#interface fa0/18
Sw1(config-if)#switchport mode access
Sw1(config-if)#switchport port-security ?
    mac-address    Secure mac address
    maximum        Max secure addresses
    violation      Security violation mode
    <cr>
Sw1(config-if)#switchport port-security
Sw1(config-if)#switchport port-security maximum ?
    <1-132>        Maximum addresses
Sw1(config-if)#switchport port-security maximum 5
Sw1(config-if)#switchport port-security mac-address ?
    H.H.H         48 bit mac address
    sticky        Configure dynamic secure addresses as sticky
Sw1(config-if)#switchport port-security mac-address sticky
Sw1(config-if)#end
```

Enables port security

Specify the # of allowed addresses

Configure secure
MAC addresses

Configuring switch security

- Sticky MAC addresses
 - enabled by using the **switchport port-security mac-address sticky** interface configuration command
 - the interface converts all the dynamic secure MAC addresses, including those that were dynamically learned before sticky learning was enabled, to sticky secure MAC addresses and **adds all sticky secure MAC addresses to the running configuration**
 - If you disable sticky learning by using the **no switchport port-security mac-address sticky** interface configuration command, the sticky secure MAC addresses remain part of the address table but are removed from the running configuration.
- When you configure sticky secure MAC addresses by using the **switchport port-security mac-address sticky mac-address** interface configuration command, these addresses are added to the address table and the running configuration. If port security is disabled, the sticky secure MAC addresses remain in the running configuration
- If you save the sticky secure MAC addresses in the configuration file, when the switch restarts or the interface shuts down, **the interface does not need to relearn these addresses**. If you do not save the sticky secure addresses, they are lost

Configuring switch security

- Security violation
 - The maximum number of secure MAC addresses has been reached on an interface, and a station whose MAC address is not in the address table attempts to access the interface
 - An address is being used on two secure interfaces in the same VLAN
- Security violation modes
 - **protect**
 - **restrict**
 - **shutdown** (default mode)

Violation Mode	Forwards Traffic	Sends Syslog Message	Displays Error Message	Increases Violation Counter	Shuts Down Port
Protect	No	No	No	No	No
Restrict	No	Yes	No	Yes	No
Shutdown	No	Yes	No	Yes	Yes

Configuring switch security

■ Configure port security violation mode

```
Sw1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Sw1(config)#interface fa0/18
Sw1(config-if)#switchport mode access
Sw1(config-if)#switchport port-security
Sw1(config-if)#switchport port-security maximum 5
Sw1(config-if)#switchport port-security mac-address sticky
Sw1(config-if)#switchport port-security violation ?
    protect    Security violation protect mode
    restrict   Security violation restrict mode
    shutdown  Security violation shutdown mode
Sw1(config-if)#switchport port-security violation restrict
Sw1(config-if)#end
```

Configuring switch security

■ Port security defaults

Feature	Default Setting
Port security	Disabled on a port.
Maximum number of secure MAC addresses	1
Violation mode	Shutdown. The port shuts down when the maximum number of secure MAC addresses is exceeded, and an SNMP trap notification is sent.
Sticky address learning	Disabled.

Configuring switch security

■ Verify port security

```
Sw1#show port-security interface fastEthernet 0/18
```

```
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
```

```
Sw1#show port-security address
```

Secure Mac Address Table

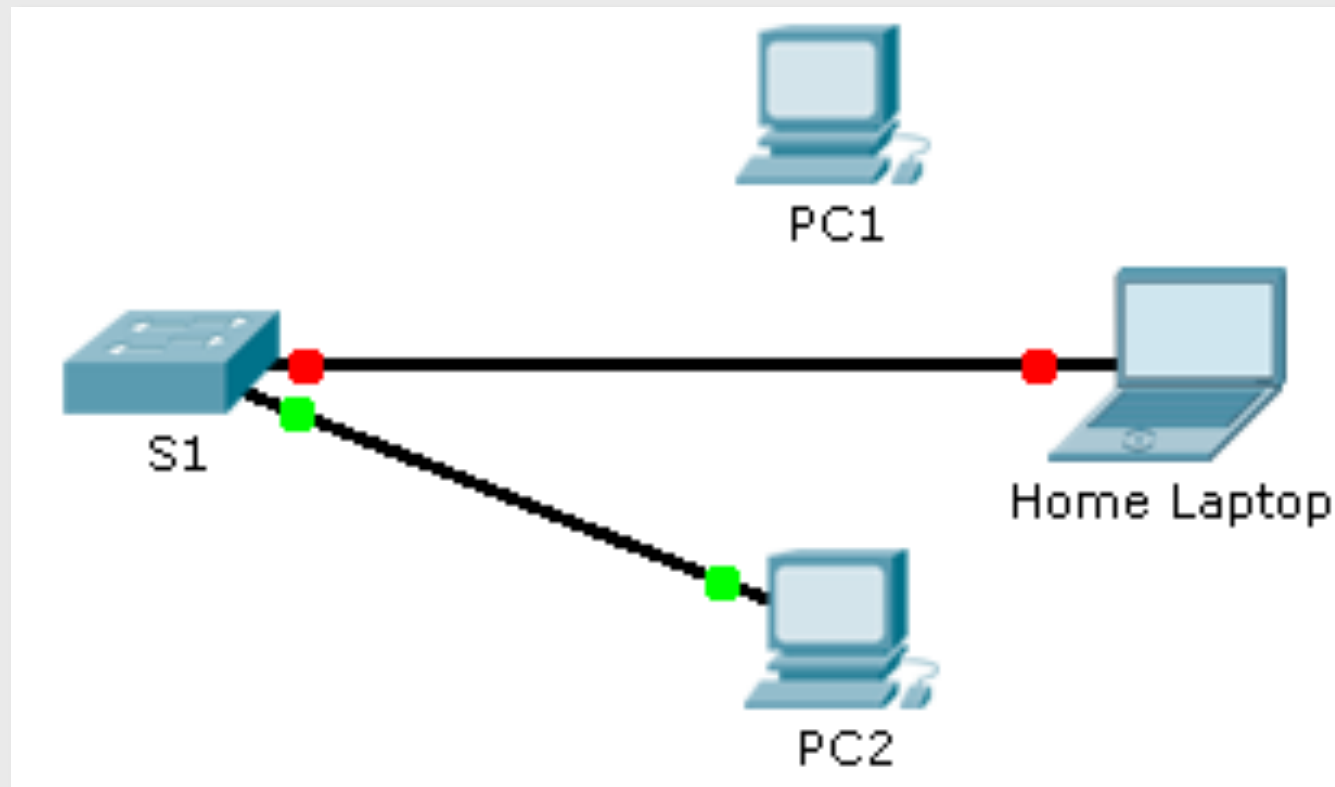
Vlan	Mac Address	Type	Ports	Remaining Age (mins)
-----	-----	----	-----	-----
99	00D0.BAED.1ACB	SecureSticky	FastEthernet0/18	-

```
-----
Total Addresses in System (excluding one mac per port) : 0
Max Addresses limit in System (excluding one mac per port) : 1024
Sw1#
```



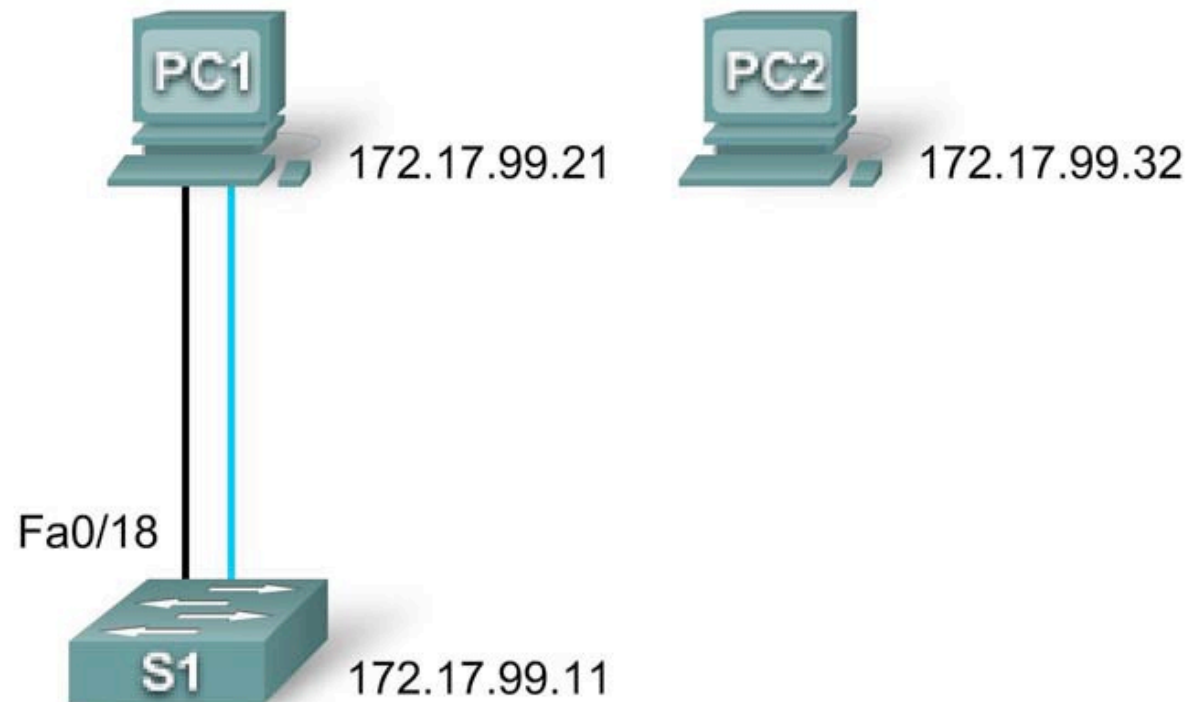
Lab activity

- Troubleshooting switch port security



Lab activity

■ Configure Switch Security



Lab activity

- Configure Switch Security



Lab activity

■ Basic Switch configuration

